

# Лабораторная работа №3

Оформление отчётов с помощью Markdown

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## **Титульный слайд**

**Дисциплина:** Архитектура компьютеров и операционные системы (раздел «Операционные системы»)

**Работа:** Лабораторная работа 3 — Markdown

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**Организация:** Российский университет дружбы народов (РУДН)

Получение навыков правильной работы с репозиториями git.

Рабочий процесс Gitflow Workflow. Будем описывать его с использованием пакета git-flow.

Выполнить работу для тестового репозитория. Преобразовать рабочий репозиторий в репозиторий с git-flow и conventional commits.

Установка из коллекции репозитория Copr  
(<https://copr.fedorainfracloud.org/coprs/elegos/gitflow/>):

Enable the copr repository `dnf copr enable elegos/gitflow` Install gitflow `dnf install gitflow`

```
ayaztgadjimuradov@ayaztgadjimuradov:~$ sudo -i
[sudo] пароль для ayaztgadjimuradov:
root@ayaztgadjimuradov:~# dnf copr enable elegos/gitflow
https://copr.fedorainfracloud.org/api_3/rpmrepo/elegos/gitflow/fedora-44/ 100% | 531.0 B/s | 560.0 B | 00m01s
Enabling a Copr repository. Please note that this repository is not part
of the main distribution, and quality may vary.

The Fedora Project does not exercise any power over the contents of
this repository beyond the rules outlined in the Copr FAQ at
<https://docs.copr.fedorainfracloud.org/user_documentation.html#what-i-can-build-in-copr>,
and packages are not held to any quality or security level.

Please do not file bug reports about these packages in Fedora
Bugzilla. In case of problems, contact the owner of this repository.
Is this ok [y/N]: y
root@ayaztgadjimuradov:~# dnf install gitflow
Обновление и загрузка репозитория:
Fedora 44 - x86_64 100% | 26.4 KiB/s | 20.0 KiB | 00m01s
Copr repo for gitflow owned by elegos 100% | 3.1 KiB/s | 2.4 KiB | 00m01s
```

На Node.js базируется программное обеспечение для семантического версионирования и общепринятых коммитов.

Fedora

```
dnf install nodejs
```

```
dnf install pnpm
```

# Настройка Node.js

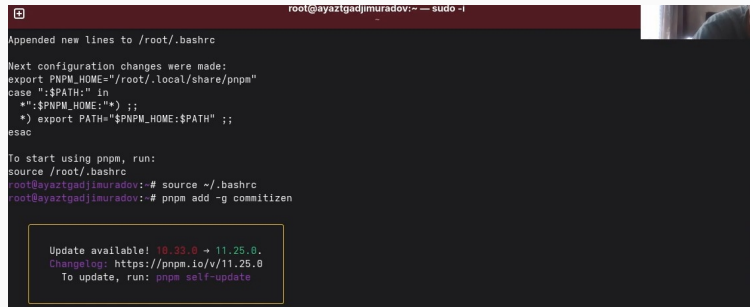
Для работы с Node.js добавим каталог с исполняемыми файлами, устанавливаемыми yarn, в переменную PATH.

Запустите:

```
pnpm setup
```

Перелогиньтесь, или выполните:

```
source ~/.bashrc
```

A terminal window with a dark background and a red title bar. The title bar contains a window icon and the text 'root@ayaztgadjimuradov:~ — sudo -i'. The terminal output shows the results of running 'pnpm setup'. It indicates that new lines were appended to '/root/.bashrc' and lists the configuration changes made, including setting 'PNPM\_HOME' and updating the 'PATH' variable. It then prompts the user to source the '.bashrc' file and shows the command 'pnpm add -g commitizen' being entered.

```
root@ayaztgadjimuradov:~ — sudo -i
Appended new lines to /root/.bashrc

Next configuration changes were made:
export PNPM_HOME="/root/.local/share/pnpm"
case ":$PATH:" in
  *:$PNPM_HOME:*) ;;
  *) export PATH="$PNPM_HOME:$PATH" ;;
esac

To start using pnpm, run:
source /root/.bashrc
root@ayaztgadjimuradov:~# source ~/.bashrc
root@ayaztgadjimuradov:~# pnpm add -g commitizen
```

```
Update available! 10.33.0 → 11.25.0.
Changelog: https://pnpm.io/v/11.25.0
To update, run: pnpm self-update
```



# Создание репозитория git

Создайте репозиторий на GitHub. Для примера назовём его git-extended. git commit -m “first commit” git remote add origin git@github.com:/git-extended.git git push -u origin master

```
root@ayaztgadjimuradov:~ — sudo -i

WARN 2 deprecated subdependencies found: glob@7.2.3, inflight@1.0.6
Packages: +148
+++++
Downloading @typescript/typescript-linux-x64@7.0.2: 9,67 MB/9,67 MB, done
Progress: resolved 167, reused 0, downloaded 148, added 148, done

/root/.local/share/pnpm/global/5:
+ commitizen 4.3.2

Done in 8.1s using pnpm v10.33.0
root@ayaztgadjimuradov:~# pnpm add -g standard-changelog
ERR_PNPM_FETCH_404 GET https://registry.npmjs.org/standard-changelog: Not Found - 404

This error happened while installing a direct dependency of /root/.local/share/pnpm/global/5
standard-changelog is not in the npm registry, or you have no permission to fetch it.

No authorization header was set for the request.
Progress: resolved 1, reused 1, downloaded 0, added 0
root@ayaztgadjimuradov:~# git commit -m "first commit"
fatal: не найден git репозиторий (или один из родительских каталогов): .git
root@ayaztgadjimuradov:~# git clone --recursive git@github.com:Sakai217/git-extended.git
Клонирование в «git-extended»...
The authenticity of host 'github.com (140.82.121.4)' can't be established.
ED25519 key fingerprint is: SHA256:+DiY3wvV6TuJJhbpZisF/zLDA0zPMSvHdKr4UvCQqu
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'github.com' (ED25519) to the list of known hosts.
warning: Похоже, что вы клонировали пустой репозиторий.
```

# Создание репозитория git

Делаем первый коммит и выкладываем на github:

```
ayaztgadjimuradov@ayaztgadjimuradov:~/git-extended — /usr/bin/node /usr/bin/pnpm init
~/git-extended

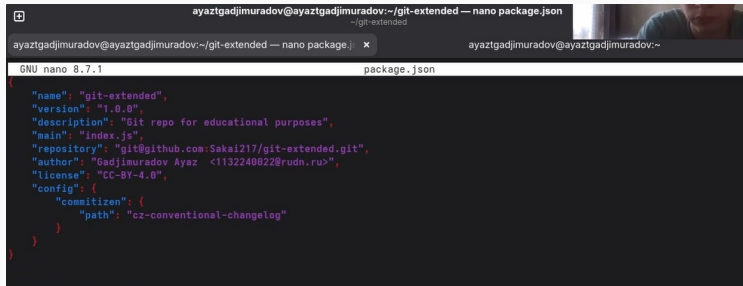
ayaztgadjimuradov@ayaztgadjimuradov:~/git-extended — /usr/bin/node /usr/bin/pnpm init x ayaztgadjimuradov@ayaztgadjimuradov:~
ayaztgadjimuradov@ayaztgadjimuradov:~/git-extended$ git remote remove origin
ayaztgadjimuradov@ayaztgadjimuradov:~/git-extended$ git remote add origin git@github.com:Sakai217/git-extended.git
ayaztgadjimuradov@ayaztgadjimuradov:~/git-extended$ git push -u origin master
git@github.com: Permission denied (publickey).
fatal: Не удалось прочесть из внешнего репозитория.

Удостоверьтесь, что у вас есть необходимые права доступа
и репозиторий существует.
ayaztgadjimuradov@ayaztgadjimuradov:~/git-extended$ git push -u origin master
Перечисление объектов: 3, готово.
Подсчет объектов: 100% (3/3), готово.
Запись объектов: 100% (3/3), 215 bytes | 215.00 KiB/s, готово.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:Sakai217/git-extended.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.
ayaztgadjimuradov@ayaztgadjimuradov:~/git-extended$ pnpm init
Wrote to /home/ayaztgadjimuradov/git-extended/package.json

{
  "name": "git-extended",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "packageManager": "pnpm@10.33.0"
}
ayaztgadjimuradov@ayaztgadjimuradov:~/git-extended$
```

# Конфигурация общепринятых коммитов

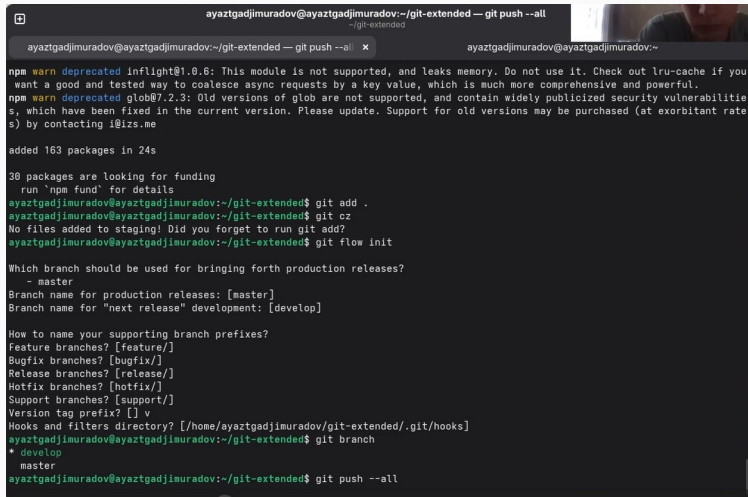
Конфигурация для пакетов Node.js npm init Сконфигурируем формат коммитов. Для этого добавим в файл package.json команду для формирования коммитов:



```
ayaztgadjimuradov@ayaztgadjimuradov:~/git-extended — nano package.json
ayaztgadjimuradov@ayaztgadjimuradov:~/git-extended — nano package.j x ayaztgadjimuradov@ayaztgadjimuradov:~
GNU nano 8.7.1 package.json
{
  "name": "git-extended",
  "version": "1.0.0",
  "description": "Git repo for educational purposes",
  "main": "index.js",
  "repository": "git@github.com:Sakai217/git-extended.git",
  "author": "Gadjimuradov Ayaz <1132240022@rudn.ru>",
  "license": "CC-BY-4.0",
  "config": {
    "commitizen": {
      "path": "cz-conventional-changelog"
    }
  }
}
```

# Конфигурация общепринятых коммитов

Добавим новые файлы, Выполним коммит, Отправим на github

A terminal window with a dark background and light green text. The window title is 'ayaztgadjimuradov@ayaztgadjimuradov:~/git-extended — git push --all'. The terminal shows several npm warnings about deprecated modules (inflight and glob), followed by a successful installation of 163 packages. Then, the user runs 'git add .' and 'git cz', which prompts for a commit message. The user then runs 'git flow init', which asks for branch names and prefixes. The user accepts the defaults for master, develop, feature, bugfix, release, hotfix, support, and version tag prefix. Finally, the user runs 'git push --all' to push the initial commit to GitHub.

```
ayaztgadjimuradov@ayaztgadjimuradov:~/git-extended — git push --all
~/git-extended
ayaztgadjimuradov@ayaztgadjimuradov:~/git-extended — git push --all x
ayaztgadjimuradov@ayaztgadjimuradov:~/git-extended:~

npm warn deprecated inflight@1.0.6: This module is not supported, and leaks memory. Do not use it. Check out lru-cache if you
want a good and tested way to coalesce async requests by a key value, which is much more comprehensive and powerful.
npm warn deprecated glob@7.2.3: Old versions of glob are not supported, and contain widely publicized security vulnerabilitie
s, which have been fixed in the current version. Please update. Support for old versions may be purchased (at exorbitant rate
s) by contacting i@izs.me

added 163 packages in 24s

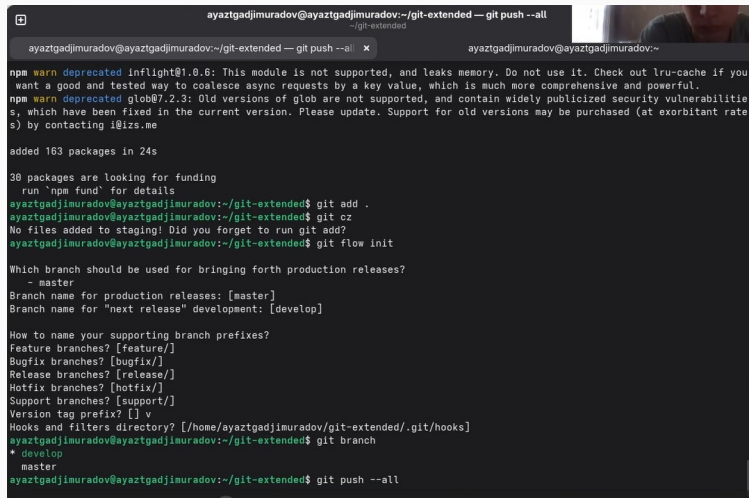
30 packages are looking for funding
  run `npm fund` for details
ayaztgadjimuradov@ayaztgadjimuradov:~/git-extended$ git add .
ayaztgadjimuradov@ayaztgadjimuradov:~/git-extended$ git cz
No files added to staging! Did you forget to run git add?
ayaztgadjimuradov@ayaztgadjimuradov:~/git-extended$ git flow init

Which branch should be used for bringing forth production releases?
- master
Branch name for production releases: [master]
Branch name for "next release" development: [develop]

How to name your supporting branch prefixes?
Feature branches? [feature/]
Bugfix branches? [bugfix/]
Release branches? [release/]
Hotfix branches? [hotfix/]
Support branches? [support/]
Version tag prefix? [] v
Hooks and filters directory? [/home/ayaztgadjimuradov/git-extended/.git/hooks]
ayaztgadjimuradov@ayaztgadjimuradov:~/git-extended$ git branch
* develop
  master
ayaztgadjimuradov@ayaztgadjimuradov:~/git-extended$ git push --all
```

# Конфигурация git-flow

Инициализируем git-flow Префикс для ярлыков установим в v.

A terminal window with a dark background and light-colored text. The window title is 'ayaztgadjimuradov@ayaztgadjimuradov:~/git-extended — git push --all'. The terminal shows the output of 'npm warn deprecated' messages, followed by 'added 163 packages in 24s'. Then, '30 packages are looking for funding' and 'run `npm fund` for details'. The user runs 'git add .' and 'git cz', which results in a message: 'No files added to staging! Did you forget to run git add?'. Then, the user runs 'git flow init'. The terminal prompts for branch names: 'Which branch should be used for bringing forth production releases?' (answer: master) and 'Branch name for "next release" development:' (answer: develop). Then, it asks for supporting branch prefixes: 'Feature branches?' (answer: feature/), 'Bugfix branches?' (answer: bugfix/), 'Release branches?' (answer: release/), 'Hotfix branches?' (answer: hotfix/), 'Support branches?' (answer: support/), and 'Version tag prefix?' (answer: v). Finally, it asks for the 'Hooks and filters directory?' (answer: [/home/ayaztgadjimuradov/git-extended/.git/hooks]). The user then runs 'git branch', showing '\* develop' and 'master'. Finally, the user runs 'git push --all'.

```
ayaztgadjimuradov@ayaztgadjimuradov:~/git-extended — git push --all
~/git-extended
ayaztgadjimuradov@ayaztgadjimuradov:~/git-extended — git push --all x
ayaztgadjimuradov@ayaztgadjimuradov:~

npm warn deprecated inflight@1.0.6: This module is not supported, and leaks memory. Do not use it. Check out lru-cache if you
want a good and tested way to coalesce async requests by a key value, which is much more comprehensive and powerful.
npm warn deprecated glob@7.2.3: Old versions of glob are not supported, and contain widely publicized security vulnerabilitie
s, which have been fixed in the current version. Please update. Support for old versions may be purchased (at exorbitant rate
s) by contacting i@izs.me

added 163 packages in 24s

30 packages are looking for funding
  run `npm fund` for details
ayaztgadjimuradov@ayaztgadjimuradov:~/git-extended$ git add .
ayaztgadjimuradov@ayaztgadjimuradov:~/git-extended$ git cz
No files added to staging! Did you forget to run git add?
ayaztgadjimuradov@ayaztgadjimuradov:~/git-extended$ git flow init

Which branch should be used for bringing forth production releases?
- master
Branch name for production releases: [master]
Branch name for "next release" development: [develop]

How to name your supporting branch prefixes?
Feature branches? [feature/]
Bugfix branches? [bugfix/]
Release branches? [release/]
Hotfix branches? [hotfix/]
Support branches? [support/]
Version tag prefix? [ ] v
Hooks and filters directory? [/home/ayaztgadjimuradov/git-extended/.git/hooks]
ayaztgadjimuradov@ayaztgadjimuradov:~/git-extended$ git branch
* develop
  master
ayaztgadjimuradov@ayaztgadjimuradov:~/git-extended$ git push --all
```

# Конфигурация git-flow

Создадим релиз с версией 1.2.3:

Создадим журнал изменений

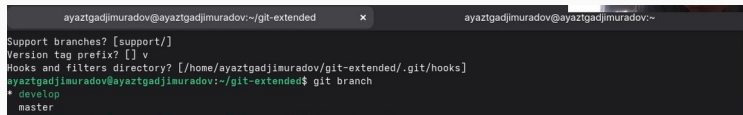
standard-changelog

Добавим журнал изменений в индекс

```
git add CHANGELOG.md
```

```
git commit -am 'chore(site): update changelog'
```

Зальём релизную ветку в основную ветку. Отправим данные на github

A terminal window with a dark background. The title bar shows two tabs: 'ayaztgadjimuradov@ayaztgadjimuradov:~/git-extended' and 'ayaztgadjimuradov@ayaztgadjimuradov:~'. The terminal content shows the output of 'git branch' command, listing 'develop' and 'master' branches.

```
Support branches? [support/]
Version tag prefix? [] v
Hooks and filters directory? [/home/ayaztgadjimuradov/git-extended/.git/hooks]
ayaztgadjimuradov@ayaztgadjimuradov:~/git-extended$ git branch
* develop
  master
```

# Создание релиза git-flow

Создадим релиз на github с комментарием из журнала изменений:  
`gh release create v1.2.3 -F CHANGELOG.md`

The screenshot displays the GitHub interface for the repository 'git-extended' owned by 'Sakai217'. At the top, navigation links include Code, Issues, Pull requests, Actions, Projects, Wiki, Security and quality, Insights, and Settings. Below the repository name, there are buttons for Pin, Watch (0), Fork (0), and Star (0). A yellow banner indicates that the 'develop' branch had recent pushes 1 minute ago, with a 'Compare & pull request' button. The main content area shows a merge commit 'Sakai217 Merge branch 'release/1.2.3'' with commit hash '231af3a' and '7 Commits'. Below this, a table lists the files included in the merge: 'CHANGELOG.md' (chore(site): update changelog, 3 minutes ago), 'README.md' (first commit, 1 hour ago), and 'package.json' (chore(site): update changelog, 3 minutes ago). A 'README' section is partially visible at the bottom. On the right sidebar, the 'About' section is empty, and the 'Releases' section shows a new release 'v1.2.3' marked as 'Latest' and published 1 minute ago. A '+ 1 release' link is visible below the release. The 'Packages' section at the bottom indicates no packages are published.